Understanding and Predicting Election Dynamics in Democracy via Multi-Agent Systems Theory

Project Proposal: Multi-Agent Systems [AI3403]

Team Name: Make Elections Great Again [MEGA]

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1 Problem Statement

Elections are, structurally, a large-scale multi-agent consensus process: voters are agents whose opinions evolve through social-network interaction, campaigns, manifestos, and news events, and the aggregate of these evolving states produces an emergent election outcome. Classical polling treats voters as static, isolated data points and cannot explain *why* opinions shift, how influence propagates, or why some electorates polarize instead of converging. **We ask: can voter opinion evolution, candidate campaigning, and constituency-level outcomes be modelled and reproduced as an emergent property of a networked multi-agent system, using consensus and bounded-confidence opinion dynamics?** We build this hierarchically - single constituency → multi-constituency (state) → national – starting with a fully working single-constituency model as our core, non-negotiable deliverable.

2 Desired End Goal (2-Month Scope)

The committed core deliverable is a *validated single-constituency simulation*; state/national extensions are explicit stretch goals attempted only if time remains.

- **Core (committed):** A single-constituency agent-based model where voter agents update opinions via a DeGroot / bounded-confidence rule, influenced by campaign/manifesto signals from candidate agents and exogenous news events; includes a turnout decision rule. Calibrated and validated against one real election dataset, reproducing realistic vote-share and polarization patterns [1, 2, 5].
- **Stretch Goal 1:** Extend to multiple constituencies with a geographic spillover term, drawing on hybrid multi-agent opinion-dynamics models over complex networks [4].
- **Stretch Goal 2:** A small synthetic multi-phase national layer demonstrating how early-phase results bias later-phase voter behaviour, informed by resilient-consensus/damage-spreading analysis of how shocks propagate through a network [3].

3 Team of Five: Individual Contributions

Beyond the core consensus/opinion-dynamics model, a real election is also shaped by two very different destabilizing forces: candidate agents can deliberately broadcast persuasive-but-misleading campaign signals, while news/event agents inject unpredictable exogenous shocks that are not necessarily aligned toward or against any candidate, but still disrupt convergence and amplify volatility. Because one channel is intentionally manipulative and the other is neutral but chaotic, and each supports multiple signal types (targeted misinformation, fabricated signals, timed shocks), three members jointly cover adversarial and disruptive agent design rather than treat it as one module.

Member	Role	Deliverable
Member A	Voter & Network Agents	Builds the baseline opinion-dynamics simulation that all other agents act on.
Member B	Malicious Candidate Agents	Designs candidate agents that emit manipulative campaign signals.
Member C	News/Event Agents	Designs neutral news agents that inject unpredictable shock events.
Member D	Adversarial Targeting	Builds shared targeting logic and measures overall manipulation impact.
Member E	Verification & Testing	Handles pipeline integration, validation, and real-world comparison.

4 Project Demonstration

We aim to demonstrate the project as a small virtual election in which individual voters are represented as agents. The voters will start with different opinions and will interact with other voters through a social network. Their opinions will change over time based on these interactions, candidate campaigns, and news events.

The basic flow of the simulation will be:

$$\boxed{\text{Voters} \rightarrow \text{Social Interactions} + \text{Campaigns/News} \rightarrow \text{Opinion Evolution} \rightarrow \text{Voting} \rightarrow \text{Election Result}}$$

The main demonstration will focus on a single constituency. We will visualize the voters and their changing opinions during the election period and show how the overall support for each candidate changes over time.

A simple dashboard will provide a mental picture of the simulation, containing:

- a visualization of the voter population and their opinions
- a plot showing how candidate support changes over time
- the effect of campaign or news events on voter opinions
- the final simulated election result

For example, the simulation may show voters initially divided between two candidates. As voters interact and respond to campaigns or news, some opinions may move closer together while others may form separate groups. At the end, these individual opinion changes will collectively produce the constituency-level election result.

The simulated result will then be compared with a real election outcome to see whether the model can reproduce realistic patterns of voter behaviour.

If the core single-constituency simulation works well, we will use the same idea as the basis for extending the model to multiple constituencies and, eventually, a larger multi-phase election.

References

[1] S. Fortunato et al., "Monte Carlo Simulations of Opinion Dynamics," arXiv. https://arxiv.org/pdf/cond-mat/0501105

[2] D. Y. Charcon and L. H. A. Monteiro, "A multi-agent system to predict the outcome of a two-round election," *Applied Mathematics and Computation*, vol. 386, 2020. https://ideas.repec.org/a/eee/apmaco/v386y2020ics0096300320304409.html

[3] S. Fortunato, "Damage Spreading and Opinion Dynamics on Scale Free Networks," arXiv. https://arxiv.org/pdf/cond-mat/0405083

[4] M. Xue and S. Roy, "Averager-copier-voter models for hybrid opinion dynamics in complex networks," arXiv. https://arxiv.org/pdf/1903.12329

[5] "Opinion sharing dynamics via a stochastic hypermatrix consensus protocol," *International Journal of Control, Automation and Systems*. https://link.springer.com/10.1007/s12555-014-0061-0